HCI 510

Research Methods II

Spring 2026

Module 3: Usability Process

Lecture 9: Writing Requirements

The UX Process

Module 3

Module 4

Discover/ Empathize

Interviews & Observations

Affinity Diagrams

Define

Personas

Scenarios

Writing requirements

Design/Ideate

Brainstorming & Ideation

Prototyping & Wireframes

Evaluate

Usability testing

A/B tests

Survey

Card Sort

Requirements

What are UX Design requirements?

Gathering UX design requirements (also called UX requirements) is a discovery process including conducting interviews, discussions, and document gathering for designers to understand what they need for a project and why.

These UX requirements form part of the project's requirements which the entire team (product, UX, engineering, business, etc.) uses to guide the product development process.

https://www.uxpin.com/studio/blog/the-beginners-guide-to-capturing-ux-requirements/

Why do we gather UX requirements?

- Most importantly, to understand users and their needs. This user research is not as thorough as the empathize or research stage of the design thinking process. It's simply a foundational understanding of the product's user experience, so UX designers know where to begin.
- To set project goals and objectives
- Understand business goals from a stakeholder's point of view and align these goals with users
- Learn about technical constraints and how they impact the project
- Eliminate ambiguity and misunderstanding

Requirements are part of the Define phase

While design decisions can sometimes inform new requirements, the initial definition and documentation of requirements must happen before significant design work (wireframing/prototyping) begins to avoid working on assumptions.

Purpose: The goal of the "Define" phase is to take research findings (from the Empathize/Discover phase) and turn them into a clear, actionable plan.

Defines what the system must do (e.g., "The user shall be able to filter search results by price").

Iterative Refinement: As you begin to design and prototype, you may discover new, more detailed requirements.
Designers as Requirements Writers: In many modern, agile teams, UX designers play a key role in the discovery/requirement-writing process, sometimes creating "design requirements" to bridge user needs with technical feasibility.

3 primary categories for requirements

Business Requirements	User Requirements	Technical Requirements
"What does the business want to achieve from this project?" An example of a business requirement might be, *"Increase new user sales conversion by 20% without affecting retention."*	*Who are the users, and what do they need from this project?* Are there any specific usability or accessibility issues designers must solve?	Functional: Outlines the product's specifications, technical capabilities, and limitations Non-functional: Describes the product's performance, such as usability, performance, data integrity, and maintenance

Business-specific requirements

Team members use stakeholder interviews and other documentation to gather and understand these business requirements.

Business requirements include:

- Project timeline
- Project scope
- Branding rules
- Various departmental goals–business, marketing, sales, customer services, etc.
- Competitors
- Stakeholder expectations

Technically-specific requirements

The requirements will answer key technical questions for design teams:

- What is the product's operating system (iOS, Android, Windows, etc.)?
- What devices will the product operate on (mobile, tablet, web, smart devices, etc.)?
- What front-end framework does the product use (React, Vue, Angular, etc.)?
- What are the product's technical limitations?

Who is responsible for gathering these requirements?

A representative from each department within the project is responsible for gathering the relevant requirements. For example:

Business: Business analyst or consultant

User: UX Lead, researcher, or designer

Technical: Tech Lead, data analyst

The above is an example of a typical project. These responsibilities will vary depending on the company, project, etc.

For example, in some cases you might be doing the user requirements and helping with business requirements.

Who is responsible for gathering these requirements?

There is no standardized step-by-step process for gathering UX requirements. The process, steps, methods, and responsibilities will differ depending on the organization and project.

There are several methods for capturing UX design requirements. The aim is to gather data relevant to your project and compile them into a single document for everyone on the project.

The first step is gathering the relevant business, user, and technical requirements. There might be some overlap in the activities team members perform to get this data. For example, surveys and interviews are relevant to business and user requirements.

How can you gather this information?

Business Requirements	User Requirements	Technical Requirements
1:1 Interviews – meeting with individual stakeholders and experts allows you to discuss specific topics in detail	User interviews	Technical stakeholder interviews
Workshops – an opportunity to discuss issues with multiple stakeholders and team members	Usability testing existing products	Systems reviews
Documentation – reviewing various business guidelines, manuals, and other documentation relevant to the project	Existing user experience research–user personas, journeys	Technical documentation and specifications
Competitor research Surveys Storyboards Moodboards	Surveys	Data analysis
	Diary studies	
	Focus groups	
	Storyboards	

User Requirements

What user requirements are not...

User requirements describe what users must be able to do, achieve, or experience — derived directly from research insights.

They are not:

- UI features
- Technical implementation details
- Design solutions
- Vague goals like “make it easy”

Research Insight	❌ Feature	✅ User Requirement
Students forget deadlines when switching between tools	“Add push notifications”	Users must be able to see all deadlines in one unified view

Step-by-Step: How to Write Effective Design Requirements

1. Understand the Project Goals

Start with the why. Every requirement should link back to a clear business or user goal.

Ask:

- What problem is this solving?
- Who are the target users?
- What is the measurable outcome?
- Keeping these visible throughout the design phase ensures alignment.

https://claritee.io/blog/how-to-write-effective-design-requirements/

Step-by-Step: How to Write Effective Design Requirements

2. Define the Scope

Define exactly what the project covers — and what it doesn't.
This keeps expectations realistic and protects the timeline.

Example:
Include homepage, product page, and checkout flow.
Exclude email templates and admin dashboard.

Step-by-Step: How to Write Effective Design Requirements

3. Identify User Needs and Personas

Good design starts with understanding real users.
Develop personas that include:

Goals and motivations
Pain points
Behavioral patterns
Example Requirement:
“The homepage should guide new users toward the product catalog within one click.”

Step-by-Step: How to Write Effective Design Requirements

4. Prioritize Features and Functionality

Not every feature matters equally.
Group them into:

Must-Haves – core functionality
Nice-to-Haves – enhancements that add polish

Example:
Must-Have: Mobile responsiveness
Nice-to-Have: Color theme customization

Step-by-Step: How to Write Effective Design Requirements

5. Write Specific, Actionable Requirements

Vague instructions create chaos. Requirements must be precise and measurable.

Clear:
“Include a persistent top navigation bar with Home, About, Services, and Contact.”

Unclear:
“Add a user-friendly menu.”

Step-by-Step: How to Write Effective Design Requirements

6. Address Technical Constraints

Every design lives inside certain limits — performance, frameworks, devices, and integrations.

Example:
“Optimize all images for a sub-3-second load time on mobile.”

Step-by-Step: How to Write Effective Design Requirements

7. Include Aesthetic and Branding Guidelines

Ensure consistency across the entire product:

Colors and typography
Visual hierarchy
Layout rhythm
Example:
“Use the primary blue (#0047FF) and white (#FFFFFF) for all call-to-action sections.”

Step-by-Step: How to Write Effective Design Requirements

8. Include Accessibility Requirements

Accessibility isn’t optional.
Define standards early so they’re built in, not patched later.

Examples:

Meet WCAG 2.1 AA
Provide alt text for images
Ensure high color contrast

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Studio #6 writing requirements

Let’s practice!

What can we take from our research findings? You’ve already done some work with your Affinity Diagrams and Personas with the Apple Watch data.

Note the following to build user requirements:

- Pain points
- Behavioral patterns
- Constraints
- Emotional drivers

Step 1: Using the sample templates below, work with your group to create requirements.

Template 1 (Clear and Strong)

The system must allow users to [accomplish goal] so that they can [achieve outcome].

Example:

The system must allow users to view all upcoming deadlines across courses in one place so that they can prioritize effectively.

Template 2 (User-Centered Framing)

Users must be able to [action] without [barrier].

Example:

Users must be able to track changes to deadlines without manually re-entering information.

Step 2: categorize each requirement as:

- Functional (what users must do)
- Informational (what users must know)
- Experiential (how it should feel)
- Accessibility-related

Practice: Why don't the first 2 examples work?

❌ "The app should send reminders."
❌ "Add a dashboard with calendar integration."
✅ "Users must be able to see deadline updates from multiple sources in one consolidated view."